DOC 1 — EVIDENCE: DATA / GRAPH / TABLE — 50 MCQs

EASY (Q1–Q10)

Q1. [Easy]

Mineral	Hardness (Mohs scale)
Talc	1
Gypsum	2
Fluorite	4
Quartz	7

A geology student is comparing the hardness of common minerals. The table shows that the hardness of fluorite on the Mohs scale is _____

Which choice most effectively uses data from the table to complete the statement?

A) 1. B) 2. C) 4. D) 7.

Q2. [Easy]

Bird species	Average wingspan (cm)
Eurasian jay	54
Common starling	38
Barn swallow	33
House sparrow	24

An ornithologist recorded the average wingspans of four bird species commonly found in suburban parks in northern Europe.

According to the table, what is the average wingspan, in centimeters, of the common starling?

A) 24 B) 33 C) 38 D) 54

Q3. [Easy]

Country	Percentage of electricity from wind power (2022)
Denmark	55.2
Ireland	33.5
Portugal	26.4
Germany	22.1

An energy analyst is examining how much of each country's electricity came from wind power in 2022. The table shows that the percentage of electricity generated from wind power in Portugal was _____

Which choice most effectively uses data from the table to complete the statement?

A) 55.2%. B) 33.5%. C) 26.4%. D) 22.1%.

Q4. [Easy]

Whale species	Estimated population
North Atlantic right whale	340
Blue whale	10,000
Humpback whale	84,000
Gray whale	27,000

Marine biologists track the estimated populations of several whale species. The species with the largest estimated population is the _____

Which choice most effectively uses data from the table to complete the statement?

A) North Atlantic right whale, with an estimated population of 340. B) blue whale, with an estimated population of 10,000. C) humpback whale, with an estimated population of 84,000. D) gray whale, with an estimated population of 27,000.

Q5. [Easy]

Telescope	Year of first light	Primary mirror diameter (meters)
Hale	1949	5.1
Keck I	1993	10.0
Gran Telescopio Canarias	2009	10.4
Vera C. Rubin Observatory	2025	8.4

An astronomy student is researching the history of large optical telescopes.

According to the table, what is the primary mirror diameter, in meters, of the Gran Telescopio Canarias?

A) 5.1 B) 8.4 C) 10.0 D) 10.4

Q6. [Easy]

City	Average January temperature (°F)	Average July temperature (°F)
Fairbanks	−7	63
Anchorage	16	58
Juneau	28	56
Sitka	33	55

A climatologist notes that among the four Alaskan cities listed, the city with the highest average January temperature is _____

Which choice most effectively uses data from the table to complete the statement?

A) Fairbanks, at −7°F. B) Anchorage, at 16°F. C) Juneau, at 28°F. D) Sitka, at 33°F.

Q7. [Easy]

Nutrient	Amount per 100 g of raw kale	Amount per 100 g of raw spinach
Vitamin C (mg)	93	28
Calcium (mg)	254	99
Iron (mg)	1.6	2.7
Fiber (g)	4.1	2.2

A nutritionist is comparing the nutrient content of kale and spinach. She notes that raw spinach contains more of one nutrient per 100 g than raw kale does: _____

Which choice most effectively uses data from the table to complete the statement?

A) vitamin C, at 28 mg compared to 93 mg. B) calcium, at 99 mg compared to 254 mg. C) iron, at 2.7 mg compared to 1.6 mg. D) fiber, at 2.2 g compared to 4.1 g.

Q8. [Easy]

Volcano	Last eruption	Elevation (m)
Mount Pelée	1932	1,397
Soufrière Hills	2010	1,050
La Grande Soufrière	1977	1,467
Mount Scenery	Unknown	887

A geologist is studying Caribbean volcanoes.

According to the table, which volcano has the highest elevation?

A) Mount Pelée, at 1,397 m B) Soufrière Hills, at 1,050 m C) La Grande Soufrière, at 1,467 m D) Mount Scenery, at 887 m

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Q9. [Easy]

Material	Thermal conductivity (W/m·K)
Copper	385
Aluminum	205
Stainless steel	16
Glass	1.0

An engineer is selecting materials for a heat exchanger. The table shows that the material with the lowest thermal conductivity is _____

Which choice most effectively uses data from the table to complete the statement?

A) copper, at 385 W/m·K. B) aluminum, at 205 W/m·K. C) stainless steel, at 16 W/m·K. D) glass, at 1.0 W/m·K.

Q10. [Easy]

Rover	Landing year	Mass (kg)
Sojourner	1997	11.5
Spirit	2004	185
Curiosity	2012	899
Perseverance	2021	1,025

A space historian is examining how the mass of Mars rovers has changed over time. For the rover that landed in 2012, for example, the mass was _____

Which choice most effectively uses data from the table to complete the statement?

A) 11.5 kg. B) 185 kg. C) 899 kg. D) 1,025 kg.

MEDIUM (Q11–Q30)

Q11. [Medium]

Crop	Yield per hectare (tonnes), 2018	Yield per hectare (tonnes), 2023
Rice	4.5	4.9
Wheat	3.4	3.3
Maize	5.9	6.7
Sorghum	1.4	1.6

Agricultural researchers tracked the yields of four staple crops over a five-year period. A student writing a report wants to identify the crop that showed the greatest absolute increase in yield per hectare between 2018 and 2023. Consulting the table, the student determines that the crop with the greatest increase was _____

Which choice most effectively uses data from the table to complete the statement?

A) rice, which increased from 4.5 to 4.9 tonnes per hectare. B) wheat, which decreased from 3.4 to 3.3 tonnes per hectare. C) maize, which increased from 5.9 to 6.7 tonnes per hectare. D) sorghum, which increased from 1.4 to 1.6 tonnes per hectare.

Q12. [Medium]

Enzyme	Optimal pH	Optimal temperature (°C)
Pepsin	2.0	37
Trypsin	8.0	37
Salivary amylase	6.8	37
Lipase	8.0	37

A biology student is comparing the environments in which digestive enzymes function most efficiently. The student notes that the two enzymes with the most similar optimal pH values are _____

Which choice most effectively uses data from the table to complete the statement?

A) pepsin and trypsin. B) trypsin and lipase. C) salivary amylase and pepsin. D) lipase and salivary amylase.

Q13. [Medium]

Year	Number of published papers on microplastics
2010	120
2013	340
2016	1,450
2019	4,200
2022	9,800

Environmental scientist Priya Nair compiled data on the growth of academic research into microplastics contamination. Nair observed that between any two consecutive data points shown, the number of published papers more than doubled, indicating accelerating scientific interest. Which choice best describes data from the table that support Nair's observation?

A) The number of papers increased from 120 in 2010 to 340 in 2013, and from 340 in 2013 to 1,450 in 2016. B) The number of papers in 2022 was 9,800, which is greater than the number in any prior year. C) The number of papers increased from 1,450 in 2016 to 4,200 in 2019, but the rate of increase slowed between 2019 and 2022. D) The total number of papers across all years is greater than 15,000.

Q14. [Medium]

Language family	Number of living languages	Percentage of world's languages
Niger-Congo	1,542	21.7
Austronesian	1,257	17.7
Trans-New Guinea	481	6.8
Sino-Tibetan	456	6.4

A linguistics student is writing a paper about the distribution of the world's languages. The student wants to emphasize that two language families together account for the greatest combined share of the world's languages. Consulting the table, the student identifies these two families as _____

Which choice most effectively uses data from the table to complete the statement?

A) Niger-Congo and Austronesian, which together account for 39.4% of the world's languages. B) Austronesian and Trans-New Guinea, which together account for 24.5% of the world's languages. C) Trans-New Guinea and Sino-Tibetan, which together account for 13.2% of the world's languages. D) Niger-Congo and Sino-Tibetan, which together account for 28.1% of the world's languages.

Q15. [Medium]

Pigment	Wavelength of peak absorption (nm)	Color absorbed
Chlorophyll a	430	Violet-blue
Chlorophyll b	455	Blue
Beta-carotene	450	Blue
Phycoerythrin	565	Green

A plant biologist is investigating the light-harvesting properties of photosynthetic pigments. The biologist notes that among the pigments listed, the one that absorbs light at the longest wavelength is _____

Which choice most effectively uses data from the table to complete the statement?

A) chlorophyll a, with a peak absorption at 430 nm. B) chlorophyll b, with a peak absorption at 455 nm. C) beta-carotene, with a peak absorption at 450 nm. D) phycoerythrin, with a peak absorption at 565 nm.

Q16. [Medium]

Museum	Annual visitors (millions), 2019	Annual visitors (millions), 2023
Louvre	9.6	8.9
National Gallery (London)	6.0	5.7
Met (New York)	6.5	5.4
Uffizi (Florence)	4.4	5.1

A cultural researcher observed that most major art museums saw declines in annual visitors between 2019 and 2023, but one museum bucked this trend. The researcher uses the table to illustrate this point by noting that _____

Which choice most effectively uses data from the table to illustrate the researcher's point?

A) the Louvre received 9.6 million visitors in 2019 but only 8.9 million in 2023, a decrease of 0.7 million. B) the Uffizi received 4.4 million visitors in 2019 and 5.1 million in 2023, an increase of 0.7 million, while the other three museums experienced declines. C) the Met saw the largest decline, dropping from 6.5 million visitors in 2019 to 5.4 million in 2023. D) the National Gallery received 6.0 million visitors in 2019, making it the second most visited museum that year.

Q17. [Medium]

Soil type	Water retention capacity (%)	Average crop yield (kg/m²)
Clay	45	3.2
Silt	38	4.1
Sandy loam	25	3.8
Sand	12	1.9

Agricultural scientist Dr. Tomás Herrera hypothesized that soils with higher water retention capacity consistently produce higher crop yields. He gathered data from experimental plots using four different soil types.

Which choice best describes data from the table that weaken Dr. Herrera's hypothesis?

A) Sand has both the lowest water retention capacity and the lowest average crop yield. B) Silt has a higher water retention capacity than sandy loam, and silt also has a higher crop yield. C) Sandy loam has a higher water retention capacity than sand and also has a higher crop yield. D) Clay has the highest water retention capacity but does not have the highest average crop yield; silt, with a lower retention capacity, yields more.

Q18. [Medium]

Asteroid	Diameter (km)	Orbital period (years)	Albedo
Pallas	512	4.6	0.16
Hygiea	434	5.6	0.07
Interamnia	332	5.4	0.07
Europa (asteroid)	303	5.5	0.06

A planetary scientist notes that among the four asteroids listed, the two with the most similar albedo values are _____

Which choice most effectively uses data from the table to complete the statement?

A) Pallas and Hygiea. B) Hygiea and Interamnia. C) Interamnia and Europa. D) Pallas and Europa.

Q19. [Medium]

Treatment group	Number of participants	Percentage reporting symptom relief
Standard dose	150	62%
Double dose	150	64%
Placebo	150	58%

Pharmacologist Reina Okamura conducted a trial of a new anti-inflammatory compound. Okamura concluded that doubling the dosage did not produce a meaningful advantage over the standard dose in terms of symptom relief.

Which choice best describes data from the table that support Okamura's conclusion?

A) Only 58% of participants in the placebo group reported symptom relief, compared to 62% in the standard dose group. B) The double dose group and the standard dose group reported similar rates of symptom relief, at 64% and 62%, respectively. C) More participants in the double dose group reported symptom relief than in the placebo group. D) Each treatment group had the same number of participants: 150.

Q20. [Medium]

Region	Forest cover (% of land area), 2000	Forest cover (% of land area), 2020
South America	51.4	46.2
Sub-Saharan Africa	29.8	25.9
Southeast Asia	49.8	46.1
Northern Europe	53.1	54.6

An environmental policy analyst is writing a summary of global deforestation trends. The analyst wants to identify the region that experienced the largest percentage-point decrease in forest cover between 2000 and 2020. Consulting the table, the analyst identifies this region as _____

Which choice most effectively uses data from the table to complete the statement?

A) South America, which experienced a decrease of 5.2 percentage points. B) Sub-Saharan Africa, which experienced a decrease of 3.9 percentage points. C) Southeast Asia, which experienced a decrease of 3.7 percentage points. D) Northern Europe, which experienced an increase of 1.5 percentage points.

Q21. [Medium]

Building material	Embodied carbon (kg CO_2 per kg)	Compressive strength (MPa)
Conventional concrete	0.13	30
Geopolymer concrete	0.06	28
Cross-laminated timber	0.05	25
Recycled steel	0.47	250

Civil engineer Amara Osei is evaluating sustainable building materials. Osei concluded that geopolymer concrete offers a better balance of lower environmental impact and structural performance compared to cross-laminated timber.

Which choice best describes data from the table that support Osei's conclusion?

A) Geopolymer concrete has lower embodied carbon than conventional concrete and similar compressive strength. B) Cross-laminated timber has the lowest embodied carbon among all four materials. C)

Geopolymer concrete has slightly higher embodied carbon than cross-laminated timber but notably higher compressive strength. D) Recycled steel has the highest compressive strength but also the highest embodied carbon.

Q22. [Medium]

Fish species	Average length (cm)	Average weight (g)	Depth range (m)
Spotted goby	6	2	1–15
Sand smelt	14	18	0–20
Lesser weever	15	50	1–50
Rock cook wrasse	18	65	1–30

A marine ecologist is researching coastal fish in the eastern Atlantic. The ecologist notes that the species with the widest depth range is _____

Which choice most effectively uses data from the table to complete the statement?

A) the spotted goby, which inhabits depths from 1 to 15 m. B) the sand smelt, which inhabits depths from 0 to 20 m. C) the lesser weever, which inhabits depths from 1 to 50 m. D) the rock cook wrasse, which inhabits depths from 1 to 30 m.

Q23. [Medium]

Novel	Year published	Approximate word count
Middlemarch	1871	316,000
Bleak House	1853	530,000
War and Peace	1869	580,000
Les Misérables	1862	360,000

A literature student is comparing the lengths of major nineteenth-century novels. The student wants to show that the two longest novels were not written in the same decade. Consulting the table, the student notes that _____

Which choice most effectively uses data from the table to accomplish the student's goal?

A) *War and Peace* (approximately 580,000 words) was published in 1869, while *Bleak House* (approximately 530,000 words) was published in 1853. B) *Les Misérables* (approximately 360,000 words) was published in 1862, and *Middlemarch* (approximately 316,000 words) was published in 1871. C) *Bleak House* (approximately 530,000 words) was published in 1853, and *Middlemarch* (approximately 316,000 words) was published in 1871. D) *War and Peace* (approximately 580,000 words) was published in 1869, while *Les Misérables* (approximately 360,000 words) was published in 1862.

Q24. [Medium]

Insulation type	R-value per inch	Cost per square foot ($)
Fiberglass batts	3.2	0.64
Cellulose (blown-in)	3.7	0.82
Spray foam (closed-cell)	6.5	1.50
Mineral wool	3.3	1.10

A homeowner is comparing insulation options. She wants the highest R-value per inch but is also concerned about cost. She notes that the option with the highest R-value per inch is spray foam, but it is also the most expensive per square foot. She considers whether mineral wool might be a reasonable compromise and observes that _____

Which choice most effectively uses data from the table to complete the homeowner's observation?

A) mineral wool has a lower R-value per inch than cellulose but costs more per square foot. B) mineral wool has a higher R-value per inch than fiberglass batts but costs less per square foot than spray foam. C) mineral wool has the same R-value per inch as fiberglass batts and costs $1.10 per square foot. D) mineral wool has the lowest R-value per inch among all four options but also one of the lowest costs.

Q25. [Medium]

River	Length (km)	Average discharge (m^3/s)
Lena	4,400	16,350
Yenisei	3,487	19,600
Ob	5,410	12,475
Amur	2,824	10,900

A geography student is examining Siberian rivers. The student wants to point out that the river with the greatest average discharge is not the longest river in the table.

Which choice most effectively uses data from the table to support the student's point?

A) The Lena is 4,400 km long and has an average discharge of 16,350 m^3/s, making it neither the longest nor the river with the greatest discharge. B) The Yenisei has the greatest average discharge at 19,600 m^3/s, yet at 3,487 km it is shorter than the Ob, which at 5,410 km is the longest river in the table. C) The Ob is the longest river in the table at 5,410 km, and the Amur is the shortest at 2,824 km with the lowest average discharge at 10,900 m^3/s. D) The Yenisei has the greatest average discharge at 19,600 m^3/s, and it is also the longest river in the table at 5,410 km.

Q26. [Medium]

Element	Melting point (°C)	Boiling point (°C)	Density (g/cm^3)
Gallium	29.8	2,204	5.91
Cesium	28.4	671	1.93
Rubidium	39.3	688	1.53
Indium	156.6	2,072	7.31

A chemistry student is writing about elements with unusual physical properties. The student notes that gallium has the largest difference between its boiling point and its melting point among the elements in the table. The data show that this difference for gallium is _____

Which choice most effectively uses data from the table to complete the statement?

A) 2,174.2°C, compared to a difference of 642.6°C for cesium. B) 2,204°C, the highest boiling point among the four elements. C) 1,915.4°C, compared to a difference of 2,174.2°C for indium. D) 29.8°C, the lowest melting point among the four elements.

Q27. [Medium]

Pollinator species	Visits per hour (native wildflower patch)	Visits per hour (ornamental garden)
Western honeybee	12	15
Bumblebee	18	7
Hoverfly	8	9
Sweat bee	14	4

Entomologist Claire Beaulieu compared pollinator activity in two adjacent habitats and concluded that native wildflower patches attract a wider range of frequent pollinators than ornamental gardens do.

Which choice best describes data from the table that support Beaulieu's conclusion?

A) In the native wildflower patch, three of the four species (bumblebee, sweat bee, and western honeybee) each made more than 10 visits per hour, whereas in the ornamental garden only the western honeybee exceeded that rate. B) Hoverflies visited the ornamental garden more frequently than the native wildflower patch, at 9 and 8 visits per hour, respectively. C) Western honeybees made the most visits per hour in the ornamental garden, at 15, compared to 12 in the native wildflower patch. D) Sweat bees made 14 visits per hour in the native wildflower patch, the second-highest rate among all species in that habitat.

Q28. [Medium]

Textile	Water usage (liters per kg)	Carbon emissions (kg CO_2 per kg)
Conventional cotton	10,000	8.0
Organic cotton	7,000	5.5
Polyester	17	9.5
Linen	6,000	3.8

A sustainability researcher is writing about trade-offs in textile production. The researcher wants to highlight that reducing water consumption does not always correspond to lower carbon emissions.

Which choice most effectively uses data from the table to illustrate the researcher's point?

A) Organic cotton uses less water than conventional cotton (7,000 vs. 10,000 liters per kg) and also produces lower carbon emissions (5.5 vs. 8.0 kg CO_2 per kg). B) Polyester uses far less water than any other textile in the table (17 liters per kg) but produces the highest carbon emissions (9.5 kg CO_2 per kg). C) Linen uses less water than conventional cotton and also has lower carbon emissions, suggesting that both metrics can decrease together. D) Conventional cotton has both high water usage and relatively high carbon emissions compared to organic cotton.

Q29. [Medium]

Dialect region	Percentage of speakers using "pop"	Percentage of speakers using "soda"	Percentage of speakers using "coke"
Upper Midwest	82	10	5
Northeast	15	75	4
Southeast	8	18	68
Pacific Northwest	12	52	6

A sociolinguist is studying regional variation in the generic terms Americans use for carbonated beverages. The sociolinguist observes that each region has a dominant term that is used by a clear majority of speakers. However, one region's dominant term is also the one with the highest usage rate across all four regions.

Which choice most effectively uses data from the table to identify that region and term?

A) The Upper Midwest, where 82% of speakers use "pop"—the highest percentage for any single term in any region. B) The Northeast, where 75% of speakers use "soda." C) The Southeast, where 68% of speakers use "coke." D) The Pacific Northwest, where 52% of speakers use "soda."

Q30. [Medium]

Planet	Orbital period (Earth days)	Average surface temperature (°C)
Kepler-442b	112	−40
Kepler-186f	130	−60
TOI-700d	37	5
Proxima Centauri b	11	−39

An astrobiologist is evaluating the habitability of confirmed exoplanets. The astrobiologist notes that the exoplanet with the shortest orbital period does not have the highest average surface temperature. Consulting the table, the astrobiologist observes that _____

Which choice most effectively uses data from the table to complete the observation?

A) Kepler-186f has the longest orbital period at 130 days and the lowest average surface temperature at −60°C. B) Kepler-442b has an orbital period of 112 days and an average surface temperature of −40°C, making it one of the colder exoplanets. C) TOI-700d has both the shortest orbital period and the highest average surface temperature among the four exoplanets. D) Proxima Centauri b has the shortest orbital period at 11 days but has an average surface temperature of −39°C, while TOI-700d, with an orbital period of 37 days, has the highest average surface temperature at 5°C.

HARD (Q31–Q50)

Q31. [Hard]

Soil amendment	Mean root length (cm), drought condition	Mean root length (cm), irrigated condition
None (control)	8.2	14.6
Biochar	12.7	15.1
Mycorrhizal fungi	15.3	15.8
Biochar + mycorrhizal fungi	16.1	16.0

Plant physiologist Kenji Watanabe investigated whether soil amendments could improve root growth under drought stress. Watanabe concluded that the benefit of soil amendments is greatest under drought conditions, as amendments substantially increase root length relative to the control under drought but produce only marginal gains when water is plentiful. _____

Which choice most effectively uses data from the table to support Watanabe's conclusion?

A) Under drought conditions, the biochar + mycorrhizal fungi treatment yielded a mean root length of 16.1 cm, compared to 8.2 cm for the control—a difference of 7.9 cm—whereas under irrigated conditions, the same treatment yielded 16.0 cm compared to 14.6 cm for the control, a difference of only 1.4 cm. B) Mycorrhizal fungi produced the longest roots under irrigated conditions at 15.8 cm, suggesting that this amendment is effective regardless of water availability. C) The biochar + mycorrhizal fungi treatment produced mean root lengths of 16.1 cm and 16.0 cm under drought and irrigated conditions, respectively, indicating that its effect is virtually the same regardless of water availability. D) Under drought conditions, the biochar treatment increased mean root length to 12.7 cm from the control's 8.2 cm, and under irrigated conditions it increased mean root length to 15.1 cm from 14.6 cm.

Q32. [Hard]

Species	Observed frequency in restored habitat	Expected frequency based on reference habitat
Prairie dropseed	0.42	0.45
Big bluestem	0.28	0.30
Wild bergamot	0.05	0.12
Pale purple coneflower	0.03	0.10

Restoration ecologist Danielle Moreau assessed the success of a tallgrass prairie restoration by comparing plant species frequencies in the restored site with those expected based on an intact reference prairie. Moreau concluded that while dominant grasses have recovered well, certain forb species remain substantially underrepresented.

Which choice best describes data from the table that support Moreau's conclusion?

A) Wild bergamot has an observed frequency of 0.05, which is less than one-third of its expected frequency of 0.12, while the two grass species exceed their expected frequencies. B) All four species have observed frequencies lower than their expected frequencies, indicating that the restoration has not been fully successful. C) Prairie dropseed has the highest observed frequency at 0.42, which is consistent with its role as a dominant grass species in tallgrass prairies. D) Prairie dropseed and big bluestem have observed frequencies (0.42 and 0.28) close to their expected values (0.45 and 0.30), but wild bergamot and pale purple coneflower have observed frequencies (0.05 and 0.03) well below their expected values (0.12 and 0.10).

Q33. [Hard]

Ice core layer	Depth (m)	Estimated age (years before present)	CO_2 concentration (ppm)	Methane concentration (ppb)
Layer A	50	2,500	280	680
Layer B	120	8,000	230	450
Layer C	250	18,000	200	390
Layer D	400	42,000	195	380

Paleoclimatologist Ingrid Solheim analyzed ice core samples to understand the relationship between greenhouse gas concentrations and geological time. Solheim concluded that CO_2 and methane concentrations have generally increased together over the period represented by the samples, with the most substantial joint increase occurring in the most recent transition.

Which choice most effectively uses data from the table to illustrate Solheim's conclusion?

A) Between Layer D (42,000 years ago) and Layer C (18,000 years ago), CO_2 rose from 195 to 200 ppm and methane rose from 380 to 390 ppb—the smallest concurrent increases between any two consecutive layers. B) Between Layer C (18,000 years ago) and Layer A (2,500 years ago), CO_2 rose from 200 to 280 ppm and methane rose from 390 to 680 ppb, and the transition from Layer B to Layer A alone accounts for increases of 50 ppm in CO_2 and 230 ppb in methane—the largest joint increase between any two consecutive layers. C) Layer A, the shallowest and most recent sample, has the highest concentrations of both CO_2 (280 ppm) and methane (680 ppb), confirming a general upward trend. D) The transition from Layer C to Layer B shows CO_2 rising by 30 ppm and methane by 60 ppb, whereas from Layer D to Layer C,

CO_2 rose by only 5 ppm and methane by 10 ppb, suggesting the rate of increase itself accelerated over time.

Q34. [Hard]

Treatment	Bacterial count (CFU/mL), Day 0	Bacterial count (CFU/mL), Day 3	Bacterial count (CFU/mL), Day 7
Silver nanoparticles	5.0×10^6	2.1×10^4	1.5×10^3
Copper nanoparticles	5.0×10^6	8.3×10^5	4.0×10^5
Zinc oxide nanoparticles	5.0×10^6	1.9×10^6	9.2×10^5
Untreated (control)	5.0×10^6	6.8×10^6	1.2×10^7

Microbiologist Fatima El-Amin tested three metallic nanoparticle treatments for their antibacterial efficacy against a waterborne pathogen. El-Amin concluded that all three treatments reduced bacterial counts relative to the control, but silver nanoparticles were the most effective, achieving a reduction of several orders of magnitude by Day 7.

Which choice most effectively uses data from the table to support El-Amin's conclusion?

A) By Day 7, the silver nanoparticle treatment reduced the bacterial count to 1.5×10^3 CFU/mL from an initial 5.0×10^6 CFU/mL—a reduction of more than three orders of magnitude—while the control group's count increased to 1.2×10^7 CFU/mL. B) By Day 3, all three treatments had lower bacterial counts than the control, but the zinc oxide treatment still had a count of 1.9×10^6 CFU/mL, nearly as high as the initial count. C) The copper nanoparticle treatment reduced the bacterial count to 4.0×10^5 CFU/mL by Day 7, a reduction of approximately one order of magnitude from the initial count. D) The control group's bacterial count increased from 5.0×10^6 to 1.2×10^7 CFU/mL over seven days, demonstrating that untreated water allowed bacterial growth.

Q35. [Hard]

Condition	Mean response time (ms), congruent trials	Mean response time (ms), incongruent trials
No distraction	340	420
Auditory distraction	355	490
Visual distraction	380	530
Dual distraction (auditory + visual)	410	610

Cognitive psychologist Lena Strand investigated how different types of distraction affect performance on a task requiring selective attention. Strand concluded that distractions increase response times in all trial types, but the effect of distractions is disproportionately larger for incongruent trials than for congruent trials.

Which choice most effectively uses data from the table to illustrate Strand's conclusion?

A) The dual distraction condition produced the highest response times for both congruent (410 ms) and incongruent (610 ms) trials, confirming that more distractions lead to slower responses overall. B) Mean response times for congruent trials ranged from 340 ms to 410 ms across conditions, an increase of 70 ms, while incongruent trial response times in the dual distraction condition were 610 ms, the highest value in the table. C) Visual distraction produced higher response times than auditory distraction for both trial types, suggesting that visual interference is inherently more disruptive. D) In the no-distraction condition, the difference between incongruent and congruent response times is 80 ms, but in the dual distraction condition this difference widens to 200 ms, indicating that distractions amplify the difficulty of incongruent trials far more than congruent trials.

Q36. [Hard]

Country	GDP per capita ($)	Mean life satisfaction (1–10 scale)
Country A	82,000	7.4
Country B	48,000	7.6
Country C	12,000	6.1
Country D	5,500	5.8
Country E	3,200	6.3

Economist Rajesh Banerjee examined the relationship between national wealth and self-reported life satisfaction. Banerjee concluded that while life satisfaction generally increases with GDP per capita, the relationship is not perfectly linear, since some lower-income countries report higher satisfaction than wealthier peers.

Which choice most effectively uses data from the table to illustrate Banerjee's conclusion?

A) Country A has the highest GDP per capita ($82,000) and a life satisfaction score of 7.4, while Country D has a GDP per capita of $5,500 and a score of 5.8, confirming that wealthier countries are always more satisfied. B) Country A has both the highest GDP per capita and a high life satisfaction score, while Country D has both a low GDP per capita and a low satisfaction score, indicating a strong positive relationship. C) Countries C, D, and E all have GDP per capita below $15,000 and life satisfaction scores below 6.5, suggesting a threshold below which satisfaction is uniformly low. D) Country B has a lower GDP per capita ($48,000) than Country A ($82,000) yet reports higher life satisfaction (7.6 vs. 7.4), and Country E, with the lowest GDP per capita ($3,200), reports higher satisfaction (6.3) than Country D ($5,500, 5.8).

Q37. [Hard]

Adhesive	Bond strength (MPa), dry	Bond strength (MPa), submerged 24 hrs	Bond strength (MPa), submerged 72 hrs
Epoxy resin	28.5	24.1	19.8
Cyanoacrylate	22.0	9.6	3.2
Polyurethane	15.4	14.8	14.1
Marine-grade silicone	8.2	6.5	4.8

Materials engineer Sofia Lindgren tested four adhesives for use in underwater construction. Lindgren concluded that while epoxy resin offers the highest initial bond strength, polyurethane is the best choice for prolonged submersion because its bond strength remains the most stable over time.

Which choice most effectively uses data from the table to support Lindgren's conclusion?

A) Epoxy resin's bond strength declined from 28.5 MPa when dry to 19.8 MPa after 72 hours of submersion—a decrease of 8.7 MPa—whereas polyurethane's strength declined from 15.4 MPa to 14.1 MPa, a decrease of only 1.3 MPa. B) Marine-grade silicone's bond strength dropped from 8.2 MPa to 4.8 MPa after 72 hours—a decrease of over 40%—making it the least stable adhesive among the four tested. C) Cyanoacrylate experienced the largest decline, dropping from 22.0 MPa to 3.2 MPa after 72 hours, making it clearly unsuitable for underwater applications. D) Polyurethane's bond strength after 72 hours of submersion (14.1 MPa) was higher than cyanoacrylate's dry bond strength, which was 22.0 MPa.

Q38. [Hard]

Neighborhood	Median household income ($)	Average commute time (min)	Green space per capita (m^2)
Westbrook	92,000	22	45
Elm Terrace	64,000	38	18
Riverside	55,000	41	52
Northgate	78,000	30	30

Urban planner Mei Chen analyzed quality-of-life indicators across four neighborhoods. Chen concluded that access to green space does not consistently track with household income: some lower-income neighborhoods have more green space per capita than wealthier ones.

Which choice most effectively uses data from the table to support Chen's conclusion?

A) Westbrook has the highest median household income ($92,000) and 45 m^2 of green space per capita, while Northgate has the second-highest income ($78,000) and 30 m^2 of green space per capita. B) Riverside has the second-lowest median household income ($55,000) but the highest green space per capita (52 m^2), exceeding Westbrook (45 m^2), which has the highest income, and Northgate (30 m^2), which has the second-highest income. C) Elm Terrace has both a lower median household income and less green space per capita than Westbrook, showing that income and green space can be correlated. D) Northgate has a higher median income ($78,000) than Riverside ($55,000) but a shorter commute time (30 min vs. 41 min), suggesting income is tied to commuting efficiency rather than green space.

Q39. [Hard]

Model	Prediction accuracy (%), training data	Prediction accuracy (%), new data	Processing time (seconds)
Linear regression	78	76	0.3
Decision tree	92	71	1.2
Random forest	95	84	8.5
Neural network	98	86	45.0

Data scientist Owen Park evaluated four machine learning models for predicting urban traffic congestion. Park concluded that the most complex model does not offer the best trade-off between predictive performance on new data and computational efficiency.

Which choice most effectively uses data from the table to illustrate Park's conclusion?

A) The neural network achieves the highest accuracy on new data at 86%, only 2 percentage points above the random forest's 84%, but requires more than five times the processing time (45.0 seconds vs. 8.5 seconds). B) The decision tree achieves 92% accuracy on training data but only 71% on new data, indicating severe overfitting. C) Linear regression has the shortest processing time at 0.3 seconds but also the lowest accuracy on new data at 76%. D) The random forest achieves 95% accuracy on training data and 84% on new data, making it the overall best model.

Q40. [Hard]

Fertilizer type	Yield (kg/plot), Year 1	Yield (kg/plot), Year 2	Yield (kg/plot), Year 3	Soil nitrogen (mg/kg), Year 3
Synthetic NPK	52	54	48	12
Compost	38	44	50	28
Green manure	35	42	49	25
No fertilizer	30	28	25	8

Agronomist Lucia Fernandez conducted a three-year trial comparing fertilizer regimes. Fernandez concluded that organic amendments initially produce lower yields than synthetic fertilizers but demonstrate an improving trajectory over time, likely because they build soil nitrogen reserves.

Which choice most effectively uses data from the table to support Fernandez's conclusion?

A) By Year 3, compost produced a yield of 50 kg/plot, which exceeded the synthetic NPK yield of 48 kg/plot, but the green manure yield of 49 kg/plot was essentially equivalent to both. B) Green manure and compost both produced lower yields than synthetic NPK in all three years, suggesting that organic amendments are consistently inferior. C) The no-fertilizer control showed declining yields from 30 to 25 kg/plot and had the lowest soil nitrogen at 8 mg/kg, confirming that some form of fertilization is necessary. D) Synthetic NPK produced the highest yield in Year 1 (52 kg/plot) but declined to 48 kg/plot by Year 3, while compost yields rose from 38 to 50 kg/plot and compost-treated soil had 28 mg/kg of nitrogen, compared to 12 mg/kg for synthetic NPK.

Q41. [Hard]

Period	Number of shipwrecks documented	Average cargo weight (tonnes)	Percentage carrying ceramics
200 BCE–100 CE	48	75	35%
100–300 CE	112	140	62%
300–500 CE	67	95	48%
500–700 CE	23	60	29%

Maritime archaeologist Nikos Andreou studied shipwreck records from the eastern Mediterranean. Andreou concluded that both the volume and the diversity of maritime trade peaked during the period of 100–300 CE and subsequently declined, suggesting a contraction in commercial activity.

Which choice most effectively uses data from the table to support Andreou's conclusion?

A) The period 200 BCE–100 CE had 48 documented shipwrecks, fewer than the 112 documented in 100–300 CE, which indicates that trade increased before the peak. B) The period 100–300 CE had the highest number of documented shipwrecks (112), the largest average cargo weight (140 tonnes), and the highest percentage of ships carrying ceramics (62%), while all three metrics decreased in subsequent periods. C) Between 300–500 CE and 500–700 CE, the number of shipwrecks dropped from 67 to 23 and average cargo weight fell from 95 to 60 tonnes, confirming a decline after 300 CE. D) The percentage of ships carrying ceramics declined from 62% in 100–300 CE to 29% in 500–700 CE, but this decline might reflect changes in cargo preferences rather than trade volume.

Q42. [Hard]

Tree species	Mean diameter at breast height (cm), Urban	Mean DBH (cm), Suburban	Mean DBH (cm), Rural
Red maple	18.4	26.1	35.8
White oak	15.2	28.7	42.3
Eastern hemlock	12.0	19.5	31.4
American sycamore	22.6	30.4	38.9

Forestry researcher Tanisha Brooks measured tree sizes across an urbanization gradient. Brooks concluded that urban environments constrain tree growth across all species but that the magnitude of this effect varies among species: some species show a larger proportional size reduction in urban settings relative to their rural counterparts.

Which choice most effectively uses data from the table to illustrate Brooks's conclusion?

A) White oak shows the largest proportional reduction: its urban mean DBH (15.2 cm) is approximately 36% of its rural mean DBH (42.3 cm), whereas American sycamore's urban mean DBH (22.6 cm) is approximately 58% of its rural value (38.9 cm). B) All four species have smaller mean DBH in urban settings than in rural settings, confirming that urbanization constrains growth uniformly across species. C) American sycamore has the largest urban mean DBH at 22.6 cm, suggesting it is the most tolerant of urban conditions. D) Eastern hemlock's urban mean DBH (12.0 cm) is the lowest of any species in any setting, indicating it is the most sensitive to urbanization.

Q43. [Hard]

Student group	Pre-test score (mean)	Post-test score (mean), Method A	Post-test score (mean), Method B
Low-performing (below 40)	32	58	49
Mid-performing (40–70)	55	72	74
High-performing (above 70)	78	85	88

Education researcher Samuel Adeyemi compared two instructional methods across student performance tiers. Adeyemi concluded that Method A is more effective for students who begin with lower baseline scores, while Method B benefits higher-performing students more.

Which choice best describes data from the table that support Adeyemi's conclusion?

A) High-performing students scored 85 under Method A and 88 under Method B, a difference of only 3 points, which may not be statistically significant. B) Mid-performing students scored higher under Method B (74) than under Method A (72), indicating that Method B is superior for all students above the lowest tier. C) Low-performing students had the largest overall gains regardless of method, suggesting that both methods are most effective for students with the most room to improve. D) Under Method A, low-performing students improved from 32 to 58 (a gain of 26 points), compared to a gain of only 17 points under Method B (32 to 49), whereas high-performing students gained more under Method B (78 to 88, a gain of 10 points) than under Method A (78 to 85, a gain of 7 points).

Q44. [Hard]

Catalyst	Reaction rate (mol/L·s), 25°C	Reaction rate (mol/L·s), 50°C	Selectivity (%)
Platinum	0.042	0.089	92
Palladium	0.038	0.081	88
Nickel	0.015	0.058	74
Iron	0.009	0.052	65

Chemical engineer Hana Kováčová evaluated four catalysts for a hydrogenation reaction. Kováčová concluded that while all catalysts show improved reaction rates at higher temperatures, the premium catalysts (platinum and palladium) offer diminishing marginal returns in rate improvement relative to their selectivity advantage.

Which choice most effectively uses data from the table to illustrate Kováčová's conclusion?

A) Palladium achieves a selectivity of 88%, close to platinum's 92%, but has a lower reaction rate at both temperatures, suggesting it is always inferior to platinum. B) Iron has the lowest reaction rate at both temperatures and the lowest selectivity, making it the least suitable catalyst in all conditions. C) All four catalysts show higher reaction rates at 50°C than at 25°C, confirming that temperature accelerates the reaction regardless of catalyst choice. D) Platinum's reaction rate roughly doubles from 25°C to 50°C (0.042 to 0.089), while nickel's rate nearly quadruples (0.015 to 0.058), yet platinum's selectivity (92%) far exceeds nickel's (74%), meaning the rate gap narrows at higher temperatures even as the selectivity gap persists.

Q45. [Hard]

Decade	Number of languages with written grammars published	Number of languages documented only via audio recordings	Number of newly identified extinct languages
1970s	85	40	12
1980s	110	95	18
1990s	145	210	35
2000s	130	420	62

Linguist Yara Boulos analyzed trends in language documentation and extinction. Boulos concluded that although documentation efforts intensified over the four decades, the rate of language extinction accelerated even faster, suggesting that documentation alone is insufficient to prevent language loss.

Which choice most effectively uses data from the table to support Boulos's conclusion?

A) The number of languages with written grammars increased from 85 in the 1970s to 145 in the 1990s but then decreased to 130 in the 2000s, indicating that documentation efforts have plateaued. B) The number of newly identified extinct languages grew from 12 in the 1970s to 62 in the 2000s—a more than fivefold increase—while the number of written grammars published grew from 85 to 130, less than doubling over the same period. C) Audio documentation increased dramatically from 40 recordings in the 1970s to 420 in the 2000s, outpacing the growth in written grammars. D) In the 2000s, 62 languages were newly identified as extinct, the highest number in any decade, while 130 written grammars were published, the second-highest number.

Q46. [Hard]

Irrigation method	Water usage (liters/hectare/day)	Crop yield (tonnes/hectare)	Soil salinity after 5 years (dS/m)
Flood irrigation	8,500	6.2	4.8
Sprinkler	5,200	6.0	3.1
Drip irrigation	3,100	6.5	1.9
Subsurface drip	2,800	6.4	1.5

Agricultural engineer Carmen Delgado evaluated irrigation methods for long-term sustainability in arid climates. Delgado concluded that drip-based methods offer comparable or higher yields to conventional methods while dramatically reducing both water consumption and the soil salinization that threatens future productivity.

Which choice most effectively uses data from the table to support Delgado's conclusion?

A) Sprinkler irrigation uses 5,200 liters/hectare/day, roughly 60% of flood irrigation's water usage, and results in lower soil salinity (3.1 vs. 4.8 dS/m), though its yield of 6.0 tonnes/hectare is the lowest among all four methods. B) Subsurface drip uses the least water (2,800 liters/hectare/day) and has the lowest soil salinity (1.5 dS/m), but its yield of 6.4 tonnes/hectare is slightly lower than drip irrigation's 6.5 tonnes/hectare. C) Drip irrigation uses 3,100 liters/hectare/day and produces the highest yield at 6.5 tonnes/hectare, while flood irrigation uses 8,500 liters/hectare/day yet produces only 6.2 tonnes/hectare and results in the highest soil salinity at 4.8 dS/m, compared to 1.9 dS/m for drip irrigation. D) Flood irrigation produces 6.2 tonnes/hectare and has the highest soil salinity at 4.8 dS/m, indicating that high water usage directly causes soil degradation.

Q47. [Hard]

Habitat type	Species richness (mean number of species)	Evenness index (0–1)	Invasive species (% of total)
Old-growth forest	42	0.85	3%
Secondary forest (30 yr)	35	0.72	8%
Plantation forest	18	0.45	22%
Urban park	24	0.60	15%

Conservation biologist Marco Reyes compared plant diversity across four habitat types. Reyes concluded that habitat disturbance reduces not only the number of species present but also the equitability of their distribution, while simultaneously increasing vulnerability to invasive species colonization.

Which choice most effectively uses data from the table to illustrate Reyes's conclusion?

A) Old-growth forest has the highest species richness (42), the highest evenness index (0.85), and the lowest percentage of invasive species (3%), while plantation forest—the most disturbed habitat—has the lowest species richness (18), the lowest evenness (0.45), and the highest proportion of invasives (22%). B) Urban parks have higher species richness (24) than plantation forests (18) but lower evenness (0.60 vs. 0.45), contradicting the claim that richness and evenness decline together. C) Secondary forests have 35 species with an evenness index of 0.72, indicating that 30 years of recovery is insufficient to restore diversity to old-growth levels. D) The invasive species percentage increases from 3% in old-growth forest to 22% in plantation forest, but this could reflect deliberate planting of non-native species rather than invasive colonization.

Q48. [Hard]

Survey year	% of respondents preferring print newspapers	% preferring online news sites	% preferring social media news	% preferring podcasts/audio
2012	42	35	15	8
2016	28	38	24	10
2020	15	32	38	15
2024	9	26	42	23

Media researcher Anton Voronov studied shifting news consumption preferences. Voronov concluded that social media has displaced print as the dominant news source, but the transition has also been accompanied by a steady rise in audio-based news consumption, suggesting that the media landscape is fragmenting rather than simply migrating from one dominant platform to another.

Which choice most effectively uses data from the table to support Voronov's conclusion?

A) Print newspaper preference fell from 42% in 2012 to 9% in 2024, while social media preference rose from 15% to 42% over the same period, indicating a simple transfer of audience from one platform to another. B) Social media preference grew from 15% to 42% between 2012 and 2024, overtaking print (which fell from 42% to 9%), but podcast/audio preference nearly tripled from 8% to 23% during the same period, while online news sites declined from 35% to 26%, showing audience dispersal across multiple platforms rather than consolidation. C) Online news sites peaked at 38% in 2016 and declined to 26% by 2024, suggesting that digital news consumption as a whole is declining. D) By 2024, the combined preference for social media (42%) and podcasts (23%) was 65%, dwarfing print (9%) and indicating that digital platforms now dominate news consumption.

Q49. [Hard]

Population	Sample size	Mean serum vitamin D (ng/mL)	Standard deviation	% classified as deficient (<20 ng/mL)
Office workers	280	22.4	8.1	38%
Outdoor laborers	310	34.7	6.3	9%
Night-shift nurses	195	18.6	9.4	52%
Retired adults (65+)	260	25.1	10.2	31%

Epidemiologist Dr. Celine Montague investigated vitamin D status across occupational groups. Montague concluded that sunlight exposure during working hours is a stronger predictor of vitamin D levels than age, noting that groups with limited daytime outdoor exposure consistently show both lower mean serum levels and higher deficiency rates.

Which choice most effectively uses data from the table to support Montague's conclusion?

A) Office workers had a mean serum vitamin D of 22.4 ng/mL, which is above the deficiency threshold of 20 ng/mL, yet 38% were classified as deficient, indicating that the mean alone does not capture the distribution of values. B) Night-shift nurses had the lowest mean serum vitamin D (18.6 ng/mL) and the highest deficiency rate (52%), confirming that nighttime work schedules are harmful to health. C) Retired adults had a higher standard deviation (10.2) than any other group, suggesting that age introduces more variability in vitamin D status than occupational factors do. D) Outdoor laborers had the highest mean vitamin D level (34.7 ng/mL) and the lowest deficiency rate (9%), while night-shift nurses and office workers—both of whom have limited daytime outdoor exposure—had lower means (18.6 and 22.4 ng/mL) and higher deficiency rates (52% and 38%), despite being younger on average than the retired adults, who had a mean of 25.1 ng/mL and a 31% deficiency rate.

Q50. [Hard]

Emission scenario	Projected global temperature increase by 2100 (°C)	Projected sea level rise (cm)	% of coral reefs at severe risk
Low emissions (SSP1)	1.5	30	10%
Moderate emissions (SSP2)	2.5	55	35%
High emissions (SSP3)	3.5	80	75%
Very high emissions (SSP5)	4.5	110	98%

Climate scientist Dr. Rajan Gupta presented a synthesis of model projections to policymakers. Gupta concluded that the relationship between emissions and ecological impact is nonlinear: even moderate increases in temperature can trigger disproportionately large increases in the proportion of coral reefs at risk, particularly in the mid-range of projected warming.

Which choice most effectively uses data from the table to illustrate Gupta's conclusion?

A) Between SSP1 and SSP2, a temperature increase of 1.0°C raises reef risk by 25 percentage points (from 10% to 35%), whereas between SSP3 and SSP5, the same 1.0°C increase raises risk by only 23 percentage points (from 75% to 98%), suggesting a roughly consistent rather than accelerating rate of change. B) Under SSP5, 98% of coral reefs are at severe risk, compared to 10% under SSP1, confirming that higher emissions lead to greater ecological damage. C) Between SSP2 and SSP3, a 1.0°C increase corresponds to a 40-percentage-point rise in reef risk (from 35% to 75%)—a larger increase than the 25-point rise associated with the equivalent temperature increase between SSP1 and SSP2 (from 10% to 35%)—demonstrating that in the critical mid-range, each degree of warming produces a disproportionately greater ecological impact. D) Sea level rise increases from 30 cm under SSP1 to 110 cm under SSP5, a roughly proportional increase that tracks with the threefold increase in temperature (1.5°C to 4.5°C), indicating a roughly linear relationship between these variables.

END OF DOC 1 — 50 QUESTIONS